

OpenADMET PXR Structure Challenge — Executive Brief

What 30 teams did, what worked, and where the remaining accuracy is

Condensed from the full 16-page analysis of all 30 submissions (30 teams × 184 ligands), the 30 published method reports, geometric validation against the released crystal structures, and an independent field analysis presented by a rank-9 participant. Prepared for the discoverybytes team (rank 18/30, LDDT-PLI 0.468). Every figure here is carried from the full report; section references point to it.

1. Goal, summary, and how the analysis was done

Goal. The challenge asked for the bound pose of 184 novel ligands in human PXR — a nuclear receptor with an unusually large, flexible pocket that binds chemically diverse xenobiotics. Our purpose in analysing it was narrower than ranking the field: we wanted to know why our own entry placed 18th of 30, and whether the deficit was a folding problem, a sampling problem, or a scoring problem.

The competition was not won by a better structure predictor. It was won by better pose selection. All 10 Tier-1 entries used the same class of generative machinery — deep-learning cofolding. What separated them was how many independent models they ran and how they chose among the resulting poses. The single largest quantity in the data is not any team's score but the accuracy that was generated and then discarded at the ranking step.

How the analysis was done, in four layers, each independently checkable:

1. **Scores.** Per-ligand LDDT-PLI for all 30 teams recovered from the challenge platform's comparison service. Recomputed team means match the published leaderboard to ≤ 0.0011 , which validates the pull.
2. **Methods.** All 30 method reports fetched and read, with generators, sampling depth and selection rule extracted into a structured schema. Four required PDF or embedded-document extraction.
3. **Geometry.** Our own 184 poses compared directly against the released PHENIX-refined crystal structures using symmetry-corrected ligand RMSD and pocket-backbone C α RMSD. Reproduces the official BiSyRMSD to 0.7%.
4. **Pose-level corroboration.** A rank-9 participant's public presentation supplied per-pose confidence statistics from a 92,000-model pool — the one layer our score-only data could not reach.

Coordinates are the binding constraint on layer 3: only 3 of 30 entries published pose sets from which a score can be reproduced, so all geometric work is on our own submission (§6, §7.3 of the full report).

2. What they did

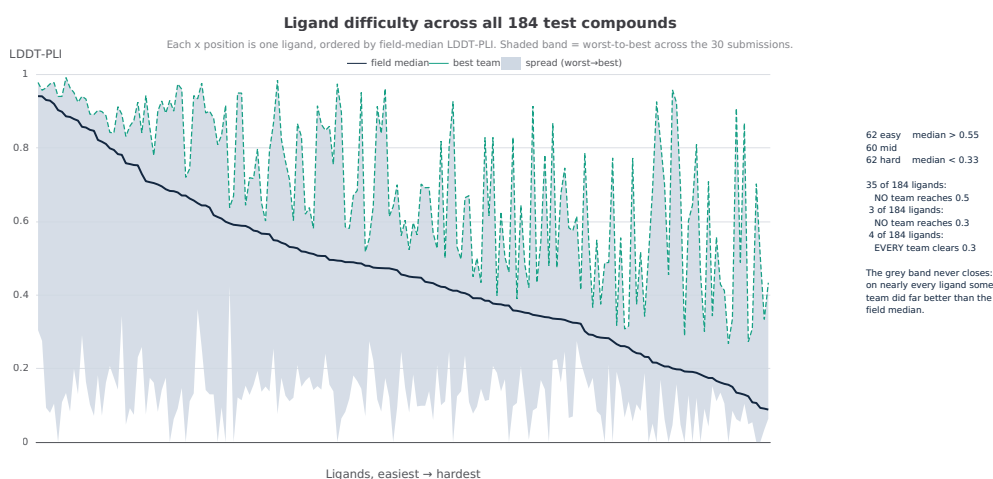
Structure generation was strikingly uniform. Of 30 entries, **19 were cofolding-only**, 9 combined cofolding with docking, 1 was docking-only (rank 27) and 1 was the organisers' template+docking baseline (rank 30). Boltz was named by 20 teams, ESMFold by 11, Protenix by 10, AlphaFold3 / OpenFold3 / Chai-1 by 7 each. Classical docking as a primary strategy finished last.

The variation that mattered was in the **selection layer** — how each team picked one pose from many:

Rank	Team	LDDT-PLI	Generators	How the final pose was chosen
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1	willvith	0.530	4 cofolders	reciprocal rank fusion of normalised ipTM + physics + PoseBusters
3	Radi	0.523	AlphaFold3 only	single-model confidence, 500 samples
5	xX-its-amit-Xx	0.517	6 cofolders + docking	cross-model z-score hybrid with tail rescue
9	dargason	0.512	6 cofolders + GNINA	hybrid threshold swap
18	discoverybytes (us)	0.468	Boltz-1 only	ipTM alone
30	DOCK baseline	0.236	UCSF DOCK	none stated

Two structural facts about the leaderboard are worth carrying: **Tier 1's 10 slots are only about 8 distinct efforts** (ranks 5–6 correlate at $r = 0.999$ and share a repository; ranks 1–2 at 0.910), and because those duplicate pairs run the same pipeline, their score difference measures the metric's own noise — **below roughly 0.02 LDDT-PLI is noise, not signal**. Tier 1 spans 0.511–0.530, so the ordering inside it is not measurable.



The difficulty landscape. All 184 ligands ordered by field-median LDDT-PLI. The shaded band spans worst-to-best across the 30 submissions and the dark line is the field median. 60% of ligands sit at or below 0.5 and only 14% above 0.75 — but the band never closes: on nearly every ligand some team scored far above the median. That vertical spread is available accuracy nobody could identify, and it is the subject of §5.

3. What worked

- **Running several architecturally different cofolders.** The number of independent models sampled correlates with rank at $p = 0.57$ ($p = 0.001$) — the strongest observational signal in the data. Diversity of inductive bias, not of random seed.
- **Selecting across models rather than within one.** The three entries that normalise confidence across cofolders are all in Tier 1. The winner fused normalised ipTM with a physics score and PoseBusters validity by rank fusion.
- **Cofolding on tractable ligands.** On the easiest third, cofolding-only pipelines mean **0.713** against 0.233 for the template+docking baseline; the Tier-1 mean on that band is 0.761 against 0.605 for ranks 11–30.
- **Getting the protein right.** Our own pocket backbone lands within 0.27–0.54 Å $C\alpha$ RMSD of the crystal structure in every pose class, including total ligand failures. Folding and template strategy were never the constraint.
- **Getting a physical read on the metric.** Recomputing symmetry-corrected RMSD for our own 184 poses makes LDDT-PLI interpretable, and locates the failure precisely:

Pose class	n (of 184)	Mean ligand RMSD	Mean LDDT-PLI	Median pocket $C\alpha$ RMSD
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Correct (< 2 Å)	39 (21%)	1.09 Å	0.822	0.27 Å
Near (2–5 Å)	84 (46%)	3.65 Å	0.479	0.38 Å
Wrong pose (5–10 Å)	59 (32%)	6.72 Å	0.235	0.42 Å
Wrong site (> 10 Å)	2 (1%)	10.98 Å	0.071	0.54 Å

The pocket backbone is essentially correct in every row, including total failures. Only 2 of 184 ligands went to the wrong site.

- **Consensus as a trust signal.** Ligands where independent pipelines agree are **1.9× more likely to be solved** than ligands where they diverge.

4. What did not work

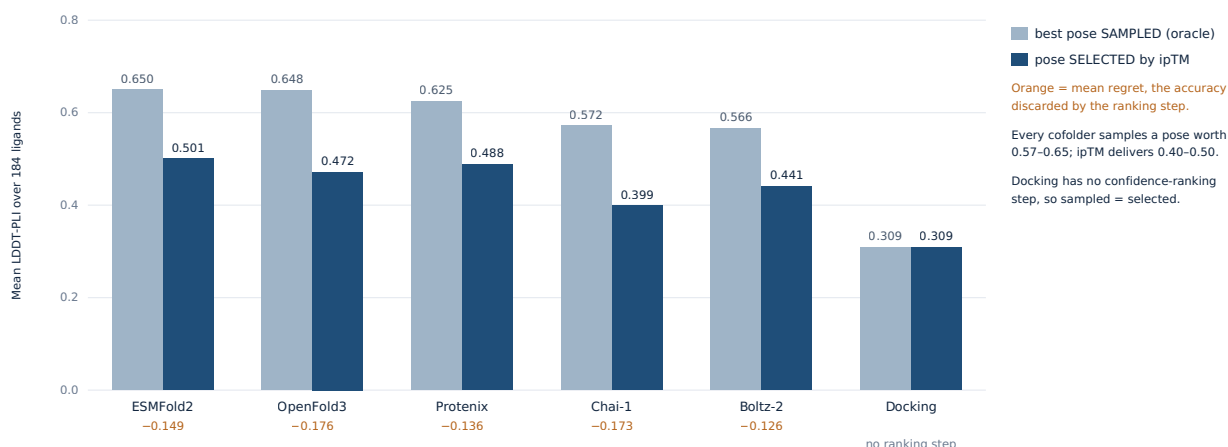
- **ipTM as a pose ranker.** This is the central failure. Within a single run, model confidence correlates with true accuracy at median Spearman $\rho = 0.13$ – 0.21 for four of five cofolders — near-useless. Our own entry used ipTM alone, and 5 of our worst poses were cases where a good pose existed in our sample pool and confidence chose wrongly.
- **Sampling harder.** Stated depth spans three orders of magnitude (5 to ~2,300 poses per ligand). Teams at both extremes reach Tier 1, and the two deepest samplers do not.
- **Every deployable selector we could test.** On the only three submissions with published coordinates: inter-team consensus recovered 5.9% of the available gap, a contact-count heuristic was worse than random, and PoseBusters validity as a selector scored –44.7% despite correlating with accuracy across 552 poses.
- **Hand-built structural priors** (from the rank-9 participant's tested-and-failed set): an apo-pocket-state prior was refuted — 47% of fragments displace the H407 side chain; multi-copy ligand modelling was wrong in both directions. Only explicit ligand protonation gave a marginal gain.
- **Predicting difficulty from chemistry.** No descriptor exceeds $|\rho| = 0.27$. Rotatable-bond count is the best of a weak field (-0.27 , $p = 0.0003$); ring count shows nothing.

5. What was unexpected

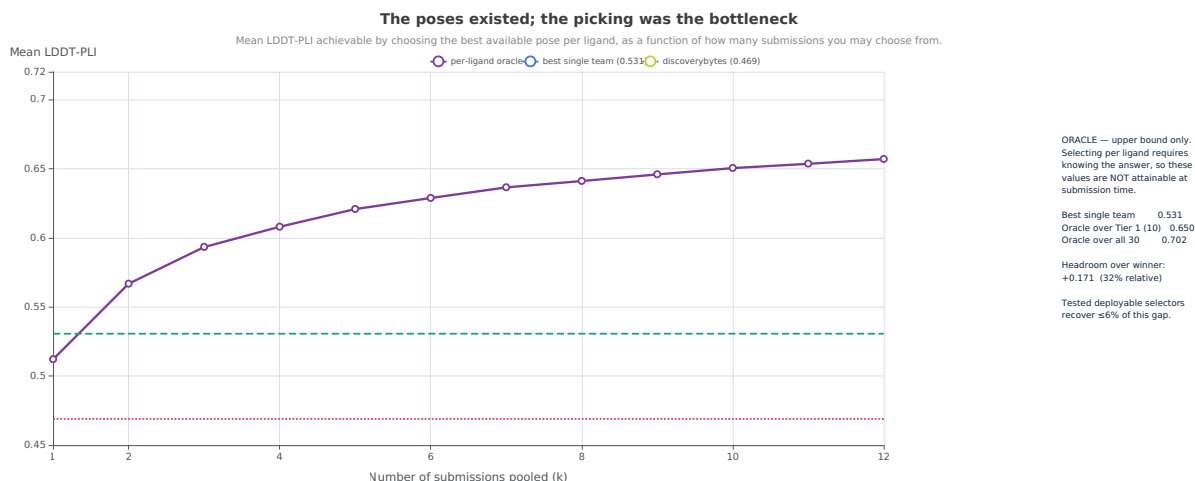
The poses that would have won the challenge were already submitted. Nobody could identify them. Choosing the best-scoring submission per ligand across the 30 entries reaches **0.702** against the winner's **0.531** — **+0.171 LDDT-PLI, or 32% relative headroom**, against the 0.019 that separates rank 1 from rank 10. This is computed from published per-ligand scores and needs no coordinates; restricting to the 22 teams with complete coverage still gives 0.685.

- **The ceiling is larger inside one pipeline than across the whole field.** The rank-9 participant's best-of-all-cofolders distribution sits ~0.24 LDDT-PLI above the best submission — near **0.77**, exceeding the 0.702 oracle over all 30 submissions. The recoverable accuracy is not a property of collective diversity; it sits inside a single well-sampled pipeline run on one consumer GPU.
- **The ranking step costs more than the choice of model.** Mean regret across five cofolders is 0.152 LDDT-PLI, larger than the 0.084 spread between the best and worst generator when each is ranked correctly.
- **Leaders win the easy ligands, not the hard ones.** The Tier-1 advantage is +0.156 on the easiest third but only +0.031 on the hardest. They are distinguished by not squandering tractable cases.
- **Last place produced the best pose on 6 of 184 ligands.** On the hardest third, the organisers' DOCK baseline ties for the most single-best poses — with a rank-14 cofolding entry, not a Tier-1 one.
- **Chai-1's confidence is anomalously informative** ($\rho = 0.49$ within-run against 0.13–0.21 for its peers), and nothing about the ligands explains why.

- **Best-pose wins are spread across 28 of 30 teams**, with the top contributor holding only 14 of 184.



The ranking step discards more accuracy than the choice of model contributors. Light bars: mean LDDT-PLI of the best pose sampled over 184 ligands. Dark bars: the pose the model's own ipTM score actually selected. Docking has no confidence-ranking stage, so its bars coincide. Values transcribed from the rank-9 participant's slides (five cofolders × 100 poses × 184 ligands, his pool, not ours).



Oracle headroom against the size of the pool you may choose from. Mean LDDT-PLI achievable by picking the best available submission per ligand, as a function of how many submissions are pooled. Three entries already capture most of the reachable gain; the curve is an upper bound requiring ground truth, not an attainable score.

6. What was learned

PXR pose prediction is **bounded by ranking, not by sampling**. Two independent measurements agree on the magnitude: a between-team oracle gap of 0.171 LDDT-PLI from published scores, and a within-pipeline regret of 0.152 from one participant's 92,000-pose pool. Our own geometry confirms the mechanism — the protein was folded correctly in every pose class, so the deficit is ligand placement and pose choice, not structure prediction.

The target's physics explains why confidence scores fail here. The pocket spans 1,100–1,800 Å³ and reshapes per ligand; two challenge ligands differing by a single methyl group adopt entirely different poses; and there are **zero hydrogen bonds** across the challenge interaction set. Binding is almost purely dispersive, leaving no polar anchor for a learned scoring function to key on.

What we would do differently, in priority order:

1. **Run 3-5 architecturally different cofolders** instead of one deeply. Pooling three entries captures roughly 60% of the achievable ceiling.

2. **Build the cross-model selector from the start, and do not trust raw ipTM.** Per-sample confidences must be logged at inference time — this cannot be retrofitted. The ranker has to add signal ipTM does not carry: physics rescoring, cross-model agreement, or validity filtering.
3. **Keep a cheap rigid-docking arm for the hard tail.** Cofolding and docking fail on different molecules.
4. **Use agreement as an uncertainty flag, not a chooser.** It predicts which ligands are tractable; it does not identify the winning pose.
5. **Audit the easy ligands first.** 53% of our own deficit came from ligands the field found tractable, including 5 catastrophic misses where a good pose sat unselected in our pool.
6. **Do not chase rank positions inside a tier.** With a noise floor near 0.02, the ordering within Tier 1 is not measurable; the 32% selection gap is where the value is.

Limitations carried from the full report. Oracle values are upper bounds by construction and require ground-truth knowledge — they are not attainable at submission time. Selector results are an $n=3$ pilot, because only 3 of 30 teams published reproducible coordinates; that pool's internal gap (0.082) is about half the field-wide figure, so those conclusions are provisional while the oracle magnitude is not. Per-generator confidence statistics and the failed-prior results are transcribed from a participant's slides, derive from his pose pool rather than ours, and could not be recomputed here. Compute-cost labels are estimates, not measurements. Method attributes are self-reported.